



10. Which of the following method is applicable only to equally spaced intervals?
 A) Lagrange's interpolation formula B) Newton's difference formula
 C) Lagrange's inverse interpolation formula D) Newton's divided difference formula
11. The n^{th} order forward difference of n^{th} degree polynomial is _____
 A) a constant B) zero
 C) a polynomial of degree 1 D) $a_0 h x$
12. First order divided difference of tabular points (7, 392) and (11, 1452) is _____
 A) 256 B) 1060
 C) 265 D) 4
13. Which of the numerical formula can be used to find $y'(x_0)$?
 A) $y'(x) = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 - \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \dots \right]$ B) $y'(x) = \frac{1}{h} \left[\Delta y_0 + \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 + \frac{1}{4} \Delta^4 y_0 + \dots \right]$
 C) $y'(x) = \frac{1}{h} \left[\Delta y_0 + \frac{1}{2} \Delta^2 y_0 - \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \dots \right]$ D) $y'(x) = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \dots \right]$
14. Given

x	0	1	2	3
y	2	4	8	14

then $y'(3) =$ _____

- A) 6 B) 7
 C) 2 D) 3
15. In Simpson's $3/8^{\text{th}}$ rule, the number of subintervals should be taken as a multiple of _____
 A) 2 B) 3
 C) 1 D) 4
16. From the table below
- | | | | |
|-----|---|---|----|
| x | 0 | 1 | 2 |
| y | 4 | 3 | 12 |
- The value of $\int_0^2 y \, dx$ using Trapezoidal rule is
 A) 15 B) 12
 C) 11 D) 9
17. A root of the equation $f(x) = 0$ lies in the interval $[a, b]$ if,
 A) $f(a)f(b) < 0$ B) $f(a)f(b) > 0$
 C) $f(a) = f(b)$ D) $f(a)f(b) = 0$
18. In Regula-falsi method, the original function $f(x)$ is approximated by a
 A) Parabola B) Tangent line
 C) Straight line D) Chord joining end points of $f(x)$
19. Which of the following represents the Poisson's equation?
 A) $\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 0$ B) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 1$
 C) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ D) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$
20. The nature of the one-dimensional heat equation is
 A) Circular B) Parabolic
 C) Hyperbolic D) Elliptic

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I****Marks BT* CO* PO***

1. a) The probability distribution of a random variable X is given by the following table

x	1	2	3	4	5	6	7
$f(x)$	k	$3k$	$5k$	$7k$	$9k$	$11k$	$13k$

Find (i) The value of k (ii) Mean and variance.

6 L*2 1 2

- b) Suppose that the following table represents the joint probability distribution of a two-dimensional random variable (X, Y)

$X \backslash Y$	1	2	3
1	1/12	1/6	0
2	0	1/9	1/5
3	1/18	1/4	2/15

Evaluate (i) the marginal distribution of X and Y .(ii) $E(X)$ and $E(Y)$.

6 L3 1 2

- c) In a certain town the duration of a shower is exponentially distributed with mean 5 minutes. What is the probability that a shower will last for
(i) less than 10 minutes
(ii) 10 minutes or more?

4 L1 2 1

2. a) Derive mean and variance of exponential distribution.

6 L3 2 2

- b) 6 coins are tossed, find the probability of getting (i) exactly 3H (ii) at most 3H (iii) at least 3H (iv) at least 1H.

6 L1 2 1

- c) Prove that $V(aX + b) = a^2V(X)$.

4 L2 1 2

3. a) In a certain factory turning out blades there is a small chance, $\frac{1}{500}$, for any blade to be defective. The blades are supplied in packets of 10. Use the P.D. to calculate the approximate number of packets containing (i) no defective blades (ii) one defective blade (iii) 2 defective blades, in a consignment of 10000 packets.

6 L2 2 2

- b) In a normal distribution, 31% of the items are under 45 and 8% are over 64. Find the mean and standard deviation of the distribution. Where $\phi(0.5) = 0.6915$ and $\phi(1.41) = 0.9207$.

6 L3 2 2

- c) The diameter of an electric cable, say X is assumed to be a continuous random variable with probability density function $f(x) = 6x(1 - x)$, for $0 < x < 1$. Obtain the cumulative distribution function of X .

4 L1 1 1

Unit – II

4. a) Show that the first difference of a polynomial of n^{th} degree is another polynomial of degree $(n - 1)$. Hence show that the n^{th} difference is a constant.

6 L2 3 2

- b) Apply Lagrange's method to find the value of x when $f(x) = 3$ from the given data.

x	4	7	10	12
$y = f(x)$	-1	1	2	4

6 L2 3 2

- c) The velocity ' v ' of a particle at distance ' s ' from a point on its path is given by the table below. Estimate the time taken to travel 60m by using Simpson's $\frac{1}{3}rd$ rule.

s (m)	0	10	20	30	40	50	60
v (m/sec)	47	58	64	65	61	52	38

4 L1 4 1

5. a) A rod is rotating in a plane. The following table gives the angle θ radians through which the rod has turned for various values of the time t (seconds)

t	0	0.2	0.4	0.6	0.8	1.0	1.2
θ	0	0.12	0.49	1.12	2.02	3.20	4.67

Calculate the angular velocity and angular acceleration of the rod when $t = 0.6 \text{ sec}$. Employ Newton's forward interpolation formula.

- b) Derive Newton-Cotes quadrature formula and hence deduce Trapezoidal rule.
- c) Find the missing term in the following table using finite differences

x	0	1	2	3	4
y	1	3	9	---	81

6. a) Using Newton's divided difference formula, evaluate $f(15)$ given

x	4	5	7	10	11	13
$f(x)$	48	100	294	900	1210	2028

- b) Compute $y'(2.2)$, given

x	1.0	1.2	1.4	1.6	1.8	2.0	2.2
y	2.72	3.32	4.06	4.96	6.05	7.37	8.97

- c) Evaluate $\int_0^1 \frac{1}{1+x^2} dx$ using Simpson's $\frac{3}{8}$ th rule by considering 7 ordinates.

Unit – III

7. a) Using Taylor's series method up to 4th power of h , solve $y' = 1 - 2xy$ at $y(0.2)$. Given $y(0) = 1$ and $h = 0.2$.
- b) Derive standard five-point formula for Laplace's equation.
- c) Find the root near x_0 , using Newton-Raphson method for $\cos x = xe^x$ with $x_0 = 3$. Carry out 2 iterations.
8. a) Apply Milne's predictor- corrector formula to compute $y(1.4)$, given that $\frac{dy}{dx} = x^2 + \frac{y}{2}$ and the data $y(1) = 2$, $y(1.1) = 2.2156$, $y(1.2) = 2.4649$ and $y(1.3) = 2.7514$.
- b) Find the solution of the parabolic equation $u_{xx} = 2u_t$ when $u(0, t) = u(4, t) = 0$ & $u(x, 0) = x(4 - x)$ taking $h = 1$, using Bendre-Schmidt recurrence relation. Find the values up to $t = 5$.
- c) Using the 4th order Runge-Kutta method find $y(0.2)$, given $y' = y - x$, $y(0) = 2$ & $h = 0.2$.

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
